

Average

✓ 1. Basic Formula

Average = Sum of observations / Number of observations

Sum = Average × Number

$$\text{Avg} = \frac{\text{Sum of elements}}{\text{No. of elements}}$$

2. Average of Natural Numbers

✓ First n Natural Numbers: Average = $(n + 1) / 2$

✓ First n Even Numbers: Average = $n + 1$

✓ First n Odd Numbers: Average = n

$$\text{No. of Avg} = \frac{\text{Sum}}{\text{Avg}}$$

$$\text{Sum of element} = \text{Avg} \times \text{No. of el.}$$

Ex. First 5 even N.N.

$$\Rightarrow \frac{2+4+6+8+10}{5} = \text{Avg}$$

$$\frac{30}{5} = 6$$

Ex. First 5 odd N.N. (Avg)

$$\frac{1+3+5+7+9}{5} = \frac{25}{5} = 5$$

3. Consecutive Numbers

✓ Average of consecutive numbers = Middle Number

Average of numbers from A to B = $(A + B) / 2$

4. Arithmetic Progression (AP)

Average = $(\text{First Term} + \text{Last Term}) / 2$

5. Adding One Observation

New Average = $(\text{Old Total} + \text{New Value}) / (\text{Old Number} + 1)$

6. Removing One Observation

New Average = $(\text{Old Total} - \text{Removed Value}) / (\text{Old Number} - 1)$

7. Average Speed

✓ Equal Distances: Average Speed = $2ab / (a + b)$

✓ Three Equal Distances: Average Speed = $3abc / (ab + bc + ca)$

8. Important Sums

- ✓ Sum of first n natural numbers = $n(n+1)/2$
- ✓ Sum of first n even numbers = $n(n+1)$
- ✓ Sum of first n odd numbers = n^2

$$n \text{ natural Number} = \frac{n+1}{2}$$

$$= \frac{\text{Sum}}{\text{No.}} \Rightarrow \frac{n(n+1)/2}{n}$$

Quick Results

Natural numbers → $(n+1)/2$

Odd numbers → n

Even numbers → $n+1$

Consecutive series → $(\text{First} + \text{Last})/2$

Average × Number = Total

Replacement → Difference/Number

Equal distance speed → $2ab/(a+b)$

Questions Type:

1. Find the average of 3, 5, 7, 8, 15?

$$\text{Avg} = \frac{\text{Sum}}{\text{No.}} = \frac{3+5+7+8+15}{5} = \frac{38}{5} = 7.6$$

2. Find the average of the given series 8,

8, 12, 16, 20, 24, 28, 32, 36, 40?

3, 6, 9, 12, 15, 18, 21, 24, 27, 30

$$\textcircled{3} (1, 2, 3, 4, 5, 6, 7, 8, 9, 10) \quad \frac{10+1}{2} = \frac{11}{2} = (5.5) \times 3 = \boxed{16.5}$$

3. Find the average of 1st 50 even numbers? $(n+1)$

$\Rightarrow$

$$50+1 = \boxed{51}$$

Ans

4. Find the average of 1st 30 odd numbers? (n)

$\Rightarrow$

5. Find the average of 1st 10 prime numbers?

$$\Rightarrow \frac{2 + 3 + 5 + 7 + 11 + 13 + 17 + 19 + 23 + 29}{10} = \frac{110 + 19}{10}$$

$$= \frac{129}{10}$$

6. Find the avg of all prime & composite no. upto 100,

$$2 - \frac{100}{2}$$

$$\frac{2+100}{2} = \frac{102}{2} = 51 \text{ Avg}$$

$$\frac{n+1}{2}$$

$$= \frac{100+1}{2}$$

$$= \frac{101}{2}$$

$$= 50.5$$

$$= 50.5$$

6. The average of first 99 even numbers is: $\frac{(n+1)}{2}$

$\Rightarrow$

100

7. The average of all the prime and composite numbers upto 100 is:

8. The average of all the perfect squares upto 100 is:

$$\Rightarrow \frac{1, 4, 9, 16, 25, 36, 49, 64, 81, 100}{10} \Rightarrow$$

$$38.5$$

9. If average of 6 consecutive even number is 31 then find the greatest numbers?

$\Rightarrow$

$n = 30$?
 $n+1 = 31$

$\text{Avg} = 31$
 $\text{No. of Terms} = 6$
 $\text{Sum} = 31 \times 6 = 186$

$x-4 + x-2 + x + x+2 + x+4 + x+6$
 $6x + 6 = 186$
 $6x = 186 - 6$
 $6x = 180$
 $x = 30$

28 30 32 34 36 38

$\Rightarrow$

10. The average age of P, Q, R is 20 years. If the average age of P and R is 16 years. What is the age of Q in years?

$\Rightarrow$

$P+Q+R = 20 \times 3 = 60$
 $P+R = 16 \times 2 = 32$
 $Q = 28$

11. Average age of Ram, Shyam and Rahul is 30 years. If average age of Rahul and Shyam is 40 Years then find the age of Ram in Years.

$\text{Sum of their age} = 30 \times 3 = 90$
 $R+S = 40 \times 2 = 80$
 $Ram = 10$

12. The average temperature of Tuesday, Wednesday and Thursday is 30°C and that of Wednesday, Thursday and Friday was 32°C. If the temperature on Tuesday is 31°C, then temperature on Friday was?

$\Rightarrow$

$T+W+Th = 30 \times 3 = 90^\circ\text{C}$
 $W+Th+Fr = 32 \times 3 = 96^\circ\text{C}$
 $T - F = -6^\circ\text{C}$
 $31^\circ - F = -6$
 $-F = -6 - 31$
 $+F = +37$

$$D = S \times T$$

$$= 25 \times \frac{1}{2} = 12.5$$

$$D = S \times T$$

$$= 40 \times \frac{1}{3} = 13.3$$

13. Amit travels 30 minutes at the speed of 25 km/hr. Further he travels 20 minutes at the speed of 40 km/hr. Find his average speed.

$$A.S = \frac{T \cdot D}{T \cdot T}$$

12.5

$$\frac{2 \times 4}{2+4} \rightarrow \text{equal distance.}$$

$$= \frac{12.5 + 13.3}{(50)}$$

$$\frac{25.8}{60}$$

$$\Rightarrow \frac{25.8 \times 60}{50}$$

$$= \frac{25.8 \times 6}{5}$$

$$\Rightarrow \frac{154.8}{5}$$

$$30.96$$

14. In the first 10 overs of a cricket game, the run rate was only 3.2. What should be the run rate in the remaining 40 overs to reach the target of 282 runs?

$$\text{Total 10 over} = 3.2 \times 10 = 32$$

$$\frac{282}{40} = 7.05$$

$$\frac{282}{-32}$$

$$250$$

15. The average rainfall in the months of January and February is 6 cm and in the months of March to June is 5 cm and July to October is 10 cm and in the November to December, it is 6 cm. The average rainfall for the whole year is:

$$J+F = 6 \times 2 = 12$$

$$M+A+M+J = 5 \times 4 = 20$$

$$J+A+S+O = 10 \times 4 = 40$$

$$N+D = 6 \times 2 = 12$$

$$\frac{84 \text{ cm}}{12} = 7 \text{ cm}$$

16. The average age of a family of 6 members 4 year ago was 25 years. Meanwhile a child was born in this family and still the average age of the whole family is same today. The present age of the child is:

$$25 \times 6 = 150 + (6 \times 4) = 150 + 24 = 174$$

$$25 \times 7 = 175$$

-1

17. The average score of Sehwag in 10 innings was 77 runs. In the 11th innings he had scored zero runs. The overall average score of Sehwag in all the 11th innings was:

$$\begin{aligned} \text{Sum} &= 10 \times 77 = 770 \\ 11^{\text{th}} &= 0 \end{aligned}$$

after 11th innings Sum = $\frac{770}{11} = 70$

18. The average weight of 20, four wheelers is 180 kg. If an old car is removed from this group of four wheelers, the new average weight decreases by 2 kg. The weight of the removed car is:

$$\Rightarrow \begin{aligned} \text{Total} &= 180 \times 20 = 3600 \text{ kg} \\ &= 178 \times 19 = 3382 \\ &\quad \underline{218} \end{aligned}$$

$$\begin{aligned} &180 - (19 \times 2) \\ &= 180 - 38 \\ &= 142 \end{aligned}$$

19. The average weight of the 5 officers of a regiment is 42 kg. If a senior officer was replaced by a new officer and thus the average increased by 500 gm, the weight of the new officer is:

$$42 + 2.5 = 44.5$$

✓

$$5 \times 42 = 210$$

$$5 \times 42.5 = 212.5$$

$$= 2.5 + 42 = 44.5$$

$$\begin{array}{r} 42.5 \times 5 \\ \hline 212.5 \\ \hline \end{array}$$

20. The average age of 10 students in a class is 20 years. If a new student is also included, then the new average age of all the student increases by 1 year. The age of the new student is: